

Kubernetes Day 8 5 June 2026

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D (main)

```
$ eksctl create cluster --name ekscluster5-june --nodegroup-name b17dng --node-type t3.micro --nodes 5 --managed
```

Resource Requests and Resource Limits

```
# 1-pod_rr_r1.yaml
---
apiVersion: v1
kind: Pod
metadata:
  name: fb-pods
  labels:
    app: fb-app
spec:
  containers:
  - name: fb-container
    image: nginx:latest
    ports:
    - containerPort: 80
    resources: #used to specify the resource requirements for the container
      requests: #requests are the minimum amount of resources that the container needs
to run
      memory: "64Mi" # 1Mi = 1024 KiB 64Mi = 64 * 1024 KiB = 65536 KiB. Mi means
Mebibyte.
      cpu: "250m" # 1 CPU = 1000m 250m = 250/1000 CPU = 0.25 CPU
    limits: #limits are the maximum amount of resources that the container can use
      memory: "216Mi"
      cpu: "500m"
```

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```
$ kubectl apply -f 1-pod_rr_r1.yaml
pod/fb-pods created
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june (main)

```
$ kubectl describe pod fb-pods
Name:          fb-pods
Namespace:     default
Priority:       0
Service Account: default
Node:          ip-192-168-30-128.ec2.internal/192.168.30.128
Start Time:    Tue, 09 Jun 2026 18:58:02 +0530
Labels:        app=fb-app
Status:        Running
IP:            192.168.30.211
IPs:
  IP: 192.168.30.211
Containers:
  fb-container:
    Container ID:
containerd://e29e5fac4f64c981544ce2c1353be2fd2d2b4aafbb6dfb23f02cf575dc4ce69e
    Image:      nginx:latest
```

Image ID:
docker.io/library/nginx@sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6

Port: 80/TCP
Host Port: 0/TCP
State: Running
Started: Tue, 09 Jun 2026 18:58:07 +0530

Limits:
cpu: 500m
memory: 216Mi
Requests:
cpu: 250m
memory: 64Mi

Events:

Type	Reason	Age	From	Message
Normal	Scheduled	110s	default-scheduler	Successfully assigned default/fb-pods to ip-192-168-30-128.ec2.internal
Normal	Pulling	110s	kubelet	spec.containers{fb-container}: Pulling image "nginx:latest"
Normal	Pulled	107s	kubelet	spec.containers{fb-container}: Successfully pulled image "nginx:latest" in 3.133s (3.133s including waiting). Image size: 63120520 bytes.
Normal	Created	107s	kubelet	spec.containers{fb-container}: Created container: fb-container
Normal	Started	106s	kubelet	spec.containers{fb-container}: Started container fb-container

env

```
# 2-postgress-pod.yaml
---
apiVersion: v1
kind: Pod
metadata:
  name: postgres-pod
  labels:
    app: postgres-app
spec:
  containers:
  - name: postgres-container
    image: postgres

    env:
    - name: POSTGRES_USER
      value: "myuser"

    - name: POSTGRES_PASSWORD # not recommended to hardcode sensitive information in code
      value: "mysecretpassword"

    - name: POSTGRES_DB
      value: "mydatabase"

    ports:
    - containerPort: 5432
```

```
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$ kubectl apply -f 2-postgres-pod.yaml
pod/postgres-pod created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june (main)
$ kubectl get all
```

NAME	READY	STATUS	RESTARTS	AGE
pod/postgres-pod	1/1	Running	0	9s

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/kubernetes	ClusterIP	10.100.0.1	<none>	443/TCP	17m

```
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```

```
$ kubectl exec -it postgres-pod -- bash
root@postgres-pod:/# env
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_SERVICE_PORT=443
HOSTNAME=postgres-pod
POSTGRES_PASSWORD=mysecretpassword
PWD=/
HOME=/root
LANG=en_US.utf8
KUBERNETES_PORT_443_TCP=tcp://10.100.0.1:443
GOSU_VERSION=1.19
PG_MAJOR=18
PG_VERSION=18.4-1.pgdg13+1
TERM=xterm
SHLVL=1
POSTGRES_USER=myuser
KUBERNETES_PORT_443_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_ADDR=10.100.0.1
PGDATA=/var/lib/postgresql/18/docker
KUBERNETES_SERVICE_HOST=10.100.0.1
KUBERNETES_PORT=tcp://10.100.0.1:443
KUBERNETES_PORT_443_TCP_PORT=443
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/lib/postgresql/18/bin
POSTGRES_DB=mydatabase
_=/usr/bin/env
root@postgres-pod:/# exit
exit
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june (main)
```

```
$ kubectl describe po postgres-pod
Name:                postgres-pod
Namespace:            default
Priority:              0
Service Account:      default
Node:                 ip-192-168-25-233.ec2.internal/192.168.25.233
Start Time:           Tue, 09 Jun 2026 23:38:32 +0530
Labels:               app=postgres-app
Annotations:          <none>
Status:               Running
IP:                   192.168.13.38
IPs:
  IP: 192.168.13.38
Containers:
  postgres-container:
    Container ID:      containerd://dc8624a208ae82a07bc86e0a8e49deb62f78ca077d23728b0aa4b93f4959ba42
    Image:             postgres
    Image ID:          docker.io/library/postgres@sha256:8ff36f3c66371cba71d20ceedccfc3de9669a68737607888c4ef0af93abe8e39
    Port:              5432/TCP
```

Host Port: 0/TCP
State: Running
Started: Tue, 09 Jun 2026 23:38:38 +0530
Ready: True
Restart Count: 0
Environment:
POSTGRES_USER: myuser
POSTGRES_PASSWORD: mysecretpassword
POSTGRES_DB: mydatabase

Events:

Type	Reason	Age	From	Message
Normal	Scheduled	96s	default-scheduler	Successfully assigned default/postgres-pod to ip-192-168-25-233.ec2.internal
Normal	Pulling	96s	kubelet	spec.containers{postgres-container}: Pulling image "postgres"
Normal	Pulled	91s	kubelet	spec.containers{postgres-container}: Successfully pulled image "postgres" in 5.393s (5.394s including waiting). Image size: 162376015 bytes.
Normal	Created	91s	kubelet	spec.containers{postgres-container}: Created container: postgres-container
Normal	Started	91s	kubelet	spec.containers{postgres-container}: Started container postgres-container

```
# 3-pod-env.yaml
---
apiVersion: v1
kind: Pod
metadata:
  name: pod-with-env
  labels:
    app: fb-app
spec:
  containers:
  - name: course-image-container
    image: nginx
    ports:
    - containerPort: 80
    env:
    - name: Course
      value: "AWS DevOps"
    - name: Batch
      value: "B17D"
    - name: Trainer
      value: "Madhu Kiran"
    - name: Duration
      value: "6 Months"
    - name: Timing
      value: "7 PM to 10 PM"
    - name: Location
      value: "Hyderabad"
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june (main)
$ kubectl apply -f 3-pod-env.yaml
pod/pod-with-env created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june (main)
$ kubectl exec -it pod-with-env -- env
```

```
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=pod-with-env
NGINX_VERSION=1.31.1
NJS_VERSION=0.9.9
NJS_RELEASE=1~trixie
ACME_VERSION=0.4.1
PKG_RELEASE=1~trixie
DYNPKG_RELEASE=1~trixie
Timing=7 PM to 10 PM
Location=Hyderabad
Course=AWS DevOps
Batch=B17D
Trainer=Madhu Kiran
Duration=6 Months
KUBERNETES_SERVICE_PORT=443
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT=tcp://10.100.0.1:443
KUBERNETES_PORT_443_TCP=tcp://10.100.0.1:443
KUBERNETES_PORT_443_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_PORT=443
KUBERNETES_PORT_443_TCP_ADDR=10.100.0.1
KUBERNETES_SERVICE_HOST=10.100.0.1
TERM=xterm
HOME=/root
```

```
#4-fb-configmap.yaml
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: fb-config
  labels:
    app: fb-app
data:
  Course: "AWS DevOps"
  Batch: "B17D"
  Trainer: "Madhu Kiran"
  Duration: "6 Months"
  Timing: "7 PM to 10 PM"
  Location: "Hyderabad"
  Institute: "Mind Circuit"
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/5_june (main)
$ kubectl apply -f 4-fb-configmap.yaml
configmap/fb-config created
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/5_june (main)
$ kubectl get all
```

NAME	READY	STATUS	RESTARTS	AGE
pod/pod-with-env	1/1	Running	0	2m42s
pod/postgres-pod	1/1	Running	0	6m32s

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/kubernetes	ClusterIP	10.100.0.1	<none>	443/TCP	24m

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/5_june (main)
$ kubectl get cm fb-config
```

NAME	DATA	AGE
fb-config	7	40s

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```
$ kubectl delete po pod-with-env
pod "pod-with-env" deleted from default namespace
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june (main)

```
$ kubectl delete po postgres-pod
pod "postgres-pod" deleted from default namespace
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june (main)

```
$ kubectl get all
NAME                                TYPE             CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
service/kubernetes                 ClusterIP         10.100.0.1      <none>           443/TCP          25m
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june (main)

```
$ kubectl get cm
NAME          DATA  AGE
fb-config     7      2m30s
kube-root-ca.crt 1      26m
```

```
# 5-fb-pod.yaml
---
apiVersion: v1
kind: Pod
metadata:
  name: fb-pod-complete
  labels:
    app: fb-app
spec:
  containers:
  - name: fb-container
    image: nginx
    ports:
    - containerPort: 80
    envFrom:
    - configMapRef:
        name: fb-config
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june (main)

```
$ kubectl apply -f 5-fb-pod.yaml
pod/fb-pod-complete created
```

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june (main)

```
$ kubectl exec -it fb-pod-complete -- env
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=fb-pod-complete
NGINX_VERSION=1.31.1
NJS_VERSION=0.9.9
NJS_RELEASE=1~trixie
ACME_VERSION=0.4.1
PKG_RELEASE=1~trixie
DYNPKG_RELEASE=1~trixie
Trainer=Madhu Kiran
Batch=B17D
Course=AWS DevOps
Duration=6 Months
Institute=Mind Circuit
```

Location=Hyderabad
Timing=7 PM to 10 PM
KUBERNETES_SERVICE_PORT=443
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT=tcp://10.100.0.1:443
KUBERNETES_PORT_443_TCP=tcp://10.100.0.1:443
KUBERNETES_PORT_443_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_PORT=443
KUBERNETES_PORT_443_TCP_ADDR=10.100.0.1
KUBERNETES_SERVICE_HOST=10.100.0.1
TERM=xterm
HOME=/root

Bhavana@LAPTOP-E0A1LIK MINGW64 ~/Downloads/Bhavana work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/5_june (main)

\$ kubectl describe po fb-pod-complete

Name: fb-pod-complete
Namespace: default
Priority: 0
Service Account: default
Node: ip-192-168-22-139.ec2.internal/192.168.22.139
Start Time: Tue, 09 Jun 2026 23:47:40 +0530
Labels: app=fb-app
Annotations: <none>
Status: Running
IP: 192.168.27.219
IPs: IP: 192.168.27.219
Containers:
fb-container:
Container ID:
containerd://7a8c69581998783e139b325c8f37a8390b482a1fc86a52019633c8e38e676008
Image: nginx
Image ID:
docker.io/library/nginx@sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6
Port: 80/TCP
Host Port: 0/TCP
State: Running
Started: Tue, 09 Jun 2026 23:47:41 +0530
Ready: True
Restart Count: 0
Environment Variables from:
fb-config ConfigMap Optional: false
Environment: <none>

Events:

Type	Reason	Age	From	Message
Normal	Scheduled	40s	default-scheduler	Successfully assigned default/fb-pod-complete to ip-192-168-22-139.ec2.internal
Normal	Pulling	40s	kubelet	spec.containers{fb-container}: Pulling image "nginx"
Normal	Pulled	39s	kubelet	spec.containers{fb-container}: Successfully pulled image "nginx" in 122ms (122ms including waiting). Image size: 63120520 bytes.
Normal	Created	39s	kubelet	spec.containers{fb-container}: Created container: fb-container
Normal	Started	39s	kubelet	spec.containers{fb-container}: started container fb-container